

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $f(x) = \sqrt[3]{x+1}$. Note that without a calculator the value $f(3)$ is difficult to determine.
 - a) Create a degree 3 Maclaurin polynomial to model the function $f(x)$.
 - b) Estimate the value of $f(3)$ using each polynomial.

- c) Create a graph of the function $f(x)$ and the approximating polynomial that you have built to model it. Feel free to use a graphing utility to assist with this.

- d) Based on your picture would expect to have overestimated or underestimated the value? How might you be able to create a better approximation for $f(3)$?

2. Create a Maclaurin polynomial of degree 2 for $f(x) = \frac{\ln(x+1)}{x+1}$ and use it to estimate the value of $\frac{2}{3}\ln(1.5)$.

3. Find the Taylor series for $f(x) = \frac{1}{(1-x)^2}$ centered at $x_0 = 0$.

4. Consider a function $f(x)$ with values of its derivative at $x = 5$ given by $f^{(n)}(5) = \frac{(-1)^n n!}{3^n (n+1)}$.

a) Find the Taylor series $T(x)$ for this function for centered at $x = 5$.

b) Find the radius of convergence and interval of convergence of this Taylor series.

5. The power series $\sum_{n=0}^{\infty} (-1)^n x^{2n+1}$ can be used to model the curve $f(x) = \frac{x}{1+x^2}$. Show how you can use the Ratio Test to determine the interval and radius of convergence for this power series.

6. Suppose that $f(x) = \sum_{n=0}^{\infty} c_n x^n$ converges for $x = -4$, but diverges for $x = 6$.

a) What can you say about the convergence of $f(x)$ at $x = 2$? What about at $x = -7$?

b) What can you say about the convergence of $\sum_{n=0}^{\infty} c_n$?

c) What can you say about the convergence of $\sum_{n=0}^{\infty} (-1)^n c_n 8^n$?

7. Find a power series whose interval of converge is

a) $(1,3)$

b) $[1,3)$

c) $(1,3]$

d) $[1,3]$